Sentiment Analysis on Pak-India Conflict Using ML and DL Models

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***Abstract*—The rapid growth of digital media has transformed the distribution of conflict-related information while also amplifying polarized, emotionally charged, and propaganda-oriented narratives. This study investigates automated analysis of news coverage related to the 2025 Pakistan–India conflict using natural language processing (NLP), machine learning (ML), deep learning (DL), and teacher–student knowledge distillation. A manually constructed dataset of 1,592 English-language news articles was collected from 13 mainstream Pakistani, Indian, and international sources during 22 April–12 May 2025. Each article was categorized into one of four content-framing classes: Official News, Government Claim, Propaganda, and Opinion. GPT-4o Mini was used as the teacher model to generate labels, rationales, and confidence scores; Llama 3.1 8B Instant was evaluated but excluded from the final analysis because it frequently generated labels outside the predefined taxonomy. Three transformer-based representations—Longformer-base-4096, BERT-base-uncased, and BigBird-RoBERTa-base—were evaluated with five ML classifiers and three DL classifiers, together with end-to-end transformer fine-tuning. Text augmentation, class-balanced weighting, and class-balanced focal loss were employed to reduce class imbalance. The best overall result was obtained by Longformer fine-tuning, achieving 82.64% accuracy and 82.75% F1-score. Among ML models, LightGBM with Longformer embeddings achieved 79.17% accuracy, while a DNN using BERT representations achieved 82.17% accuracy, providing a computationally efficient alternative to full transformer fine-tuning. The results demonstrate the potential of region-specific and context-aware models for automated monitoring of politically sensitive conflict discourse.**

***Index Terms*—Pakistan–India conflict, sentiment analysis, content classification, NLP, machine learning, deep learning, knowledge distillation, transformer, Longformer, misinformation.**

I. Introduction

The production, exchange, and consumption of information have changed substantially in the digital era. Online news platforms, social media networks, and messaging applications provide rapid access to political and social information and generate large volumes of textual material. During political unrest and military conflict, this information environment can contain independent reporting, official statements, government claims, opinion, and propaganda. Automated analysis is therefore important for understanding the framing and sentiment of rapidly changing conflict-related discourse.

Manual annotation is labor-intensive and difficult to scale when large numbers of articles are generated during a rapidly evolving crisis. NLP, ML, and DL methods provide mechanisms for processing such information at scale. Transformer-based language models can capture contextual relationships, while conventional and neural classifiers can provide computationally efficient alternatives for downstream classification.

Conflict-related analysis presents additional challenges. General-purpose sentiment and text-classification systems are frequently developed using global or Western-centric datasets and may not adequately represent South Asian geopolitical discourse. Linguistic diversity, culturally sensitive terminology, code-mixing, sarcasm, political rhetoric, and limited high-quality labeled data further complicate analysis. This study therefore focuses on English-language news coverage of the 2025 Pakistan–India conflict and investigates a region-specific framework.

The study uses a teacher–student knowledge-distillation strategy. LLMs are used to provide pseudo-labels, rationales, and confidence information, while smaller ML and DL models learn to reproduce the resulting classification at lower computational cost. The work evaluates five ML classifiers, three DL architectures, and three transformer representations.

A. Problem Statement

Public discourse surrounding the 2025 Pakistan–India crisis changed rapidly under a continuous flow of news updates, viewpoints, and conflict-related information. Conventional manual analysis is too slow and labor-intensive to provide timely insight. An automated, context-aware, and region-specific framework is therefore required to classify and analyze conflict-related news content.

B. Research Questions

The study addresses the following questions:

1) How can a teacher–student knowledge-distillation framework improve the effectiveness and interpretability of classification for Pakistan–India conflict content?
2) Can a region-specific system provide effective classification of public discourse emerging from the Pakistan–India information environment?
3) How accurately can distilled student models classify rapidly changing conflict-related information under changing contextual conditions?

C. Research Gap

The study identifies four major gaps: (1) limited conflict-specific datasets for the 2025 Pakistan–India context; (2) dependence on large manually labeled datasets; (3) inadequate contextual reasoning for geopolitical and propaganda-driven narratives; and (4) limited use of LLM-generated soft labels and rationales for training smaller models.

II. Related Work

Recent research has demonstrated the effectiveness of transformer models, knowledge distillation, multimodal learning, and ensemble methods for misinformation and news classification. Chen *et al.* investigated external knowledge distillation from LLMs for text-centric fake-news detection, while other studies explored GPT-based reasoning signals and knowledge distillation for misinformation detection [2], [6]. Indian and Pakistani datasets have also been developed for regional fake-news detection [3], [20]. Multimodal approaches have combined textual and visual representations, while graph-based approaches have modeled propagation information [4], [12].

Research specifically related to South Asian conflict discourse remains comparatively limited. Kazmi *et al.* investigated misinformation concerning the Kashmir issue using a relatively small collection of English tweets [5]. Urdu fake-news classification has been addressed using pretrained multilingual ensembles [13]. Other work has evaluated recurrent, ensemble, transformer, and transfer-learning approaches across general misinformation datasets [8], [10], [15].

Knowledge distillation is particularly relevant to the present study because it can transfer useful reasoning signals from large teacher models to computationally smaller students. Multi-teacher and multi-modal distillation approaches have also shown promise in classification and out-of-context news detection [7], [17], [19]. The present work applies this general idea to conflict-related mainstream news and evaluates both conventional ML and DL student models.

III. Dataset and Methodology

A. Research Design

The proposed architecture contains four stages:

1) construction and preparation of the conflict-news dataset;
2) knowledge generation using an LLM teacher;
3) knowledge transfer and student-model training; and
4) evaluation and comparative performance analysis.

B. Dataset

The dataset contains 1,592 updates/articles collected from 13 mainstream media sources. The temporal scope is 22 April 2025 to 12 May 2025, covering major escalations, diplomatic developments, and subsequent media coverage. Articles were selected from both print and digital editions.

The sources were divided into three groups:

- **Pakistan:** The News, The Nation, Dawn News, Pakistan Today, and The Daily Times.
- **India:** The Indian Express, The Hindu, Hindustan Times, Times of India, and Delhi Times.
- **International:** The Guardian, Washington Post, and New York Times.

Relevant articles were identified using conflict-related keywords including Pahalgam, Pakistan–India War, Operation Sindoor, Operation Bunyan-ul-Marsoos, Kashmir, and Rafale. Where newspaper archives were available only as scanned pages, OCR was used followed by manual correction. Four principal fields were retained: source, title, content, and publication date.

C. Preprocessing

The preprocessing pipeline included column selection, missing-value removal, duplicate removal, date standardization, text cleaning, normalization, stopword removal, lemmatization, lower-case conversion, and removal of very short words. Regular expressions were used to remove URLs, HTML elements, special characters, punctuation, numerical values, and unnecessary whitespace. Important acronyms and domain-specific entities such as US, UK, UN, NATO, and country names were preserved. Title and article content were cleaned separately and subsequently combined.

D. Annotation Taxonomy

Each article was assigned to one of four mutually exclusive classes:

1) Official News
2) Government Claim
3) Propaganda
4) Opinion

E. Teacher Models

Two LLMs were initially evaluated: GPT-4o Mini and Llama 3.1 8B Instant. Both are transformer-based decoder architectures that use self-attention to model contextual relationships. The basic attention operation can be expressed as

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \qquad (1)$$

GPT-4o Mini was selected as the final teacher because it adhered to the four-class taxonomy. For every article, the teacher generated:

- a classification label;
- a concise rationale describing framing, tone, language, and contextual evidence; and
- a confidence score.

Llama annotations were excluded because the model generated additional labels such as “News,” “News Report,” and “Eyewitness Account,” which were not part of the predefined taxonomy. The final GPT annotation distribution was 720 Official News, 392 Government Claim, 258 Propaganda, and 222 Opinion.

F. Class Balancing

A hybrid strategy was used to reduce class imbalance. Classes with fewer than 720 samples were augmented using WordNet-based synonym replacement with probability $p = 0.25$. When synonym augmentation was unavailable, random word deletion or duplication was used as a fallback.

Class-balanced weighting was then applied using the effective-number formulation:

$$E_i = \frac{1 - \beta^{n_i}}{1 - \beta}, \tag{2}$$

where n_i is the number of samples in class i and $\beta = 0.9999$. The resulting class weight was

$$w_i = \frac{1 - \beta}{1 - \beta^{n_i}}. \tag{3}$$

Finally, class-balanced focal loss was employed:

$$L = -w_t(1 - p_t)^{\gamma} \log(p_t), \tag{4}$$

where $\gamma = 2$, p_t is the predicted probability of the true class, and w_t is the normalized class-balanced weight.

G. Transformer Representations

Three transformer-based representations were evaluated.

Longformer-base-4096 uses local sliding-window and global attention and can process sequences of up to 4,096 tokens. It was selected for its ability to preserve long-range information in lengthy news articles.

BERT-base-uncased contains 12 transformer encoder layers, 12 attention heads, and a hidden size of 768, with approximately 110 million parameters. WordPiece tokenization was used.

BigBird-RoBERTa-base uses sparse global, local, and random attention mechanisms. For a fair experimental comparison, the study limited its maximum sequence length to 512 tokens.

The common transformer training configuration used AdamW with a learning rate of 2×10^{-5}, weight decay of 0.01, a linear learning-rate scheduler with 10% warm-up, a maximum of 50 epochs, and early stopping with patience of 10 epochs. Automatic mixed precision and gradient accumulation were used where required.

H. Student Models

Five ML classifiers were trained over transformer embeddings: Logistic Regression, Linear SVM, Random Forest, XGBoost, and LightGBM. Three DL classifiers were also evaluated: MLP, DNN, and BiLSTM.

The MLP contains two hidden layers with 512 and 256 neurons, ReLU activations, and dropout of 0.3. The DNN contains hidden layers of 1024, 512, and 256 neurons with batch normalization, ReLU activation, and dropout rates of 0.4, 0.4, and 0.3. The BiLSTM uses a sequence length of 256, 300-dimensional trainable embeddings, and a bidirectional LSTM with 256 hidden units in each direction.

The principal ML hyperparameters are summarized in Table I.

TABLE I
ML CLASSIFIER HYPERPARAMETERS

Model	Parameter	Value
Logistic Regression	max iterations	500
Linear SVM	max iterations	5000
Random Forest	estimators	300
XGBoost	estimators	300
XGBoost	max depth	6
XGBoost	learning rate	0.05
LightGBM	estimators	300
LightGBM	learning rate	0.05

I. Evaluation Metrics

Performance was evaluated using accuracy, precision, recall, and F1-score. Precision and recall are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}. \tag{5}$$

The F1-score is

$$F1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{6}$$

IV. EXPERIMENTAL RESULTS

A. Transformer Fine-Tuning

Longformer-base-4096 produced the strongest end-to-end fine-tuning result with 82.64% accuracy, 83.46% precision, 82.64% recall, and 82.75% F1-score. BERT achieved 81.02% accuracy and 81.14% F1-score. BigBird-RoBERTa achieved 72.45% accuracy and 72.26% F1-score.

The stronger Longformer result is attributed to its ability to model local and long-range dependencies in lengthy conflict-related articles containing quotations, background information, and multi-paragraph narratives.

B. Machine Learning Models

Table II summarizes the principal results. LightGBM was the best ML classifier, achieving 79.17% accuracy and 79.16% F1-score with Longformer embeddings. XGBoost reached 77.55% accuracy, while Random Forest achieved 76.16%. Linear models performed less strongly on Longformer features.

C. Deep Learning Models

The DNN was the strongest embedding-based classifier. With BERT embeddings it achieved 82.17% accuracy and 82.29% F1-score, closely matching full Longformer fine-tuning while requiring substantially less computational effort. The DNN achieved 81.94% accuracy with Longformer and 80.78% with BigBird.

MLP achieved its best result with BigBird embeddings at 75.92% accuracy. BiLSTM achieved its best result with BERT embeddings at 76.62% accuracy. The relatively modest BiLSTM improvement suggests that transformer embeddings already encode much of the sequential contextual information that recurrent models are intended to capture.

TABLE II
PERFORMANCE COMPARISON ACROSS TRANSFORMER REPRESENTATIONS

	Longformer				BigBird				BERT			
Model	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1
Fine-tuning	82.64	83.46	82.64	82.75	72.45	74.11	72.45	72.26	81.02	81.52	81.02	81.14
LR	63.66	64.11	63.66	63.71	73.84	73.90	73.84	73.86	69.68	69.68	69.68	69.67
SVM	67.59	67.76	67.59	67.58	71.99	72.15	71.99	72.04	68.52	68.84	68.52	68.53
RF	76.16	76.78	76.16	76.32	74.54	75.12	74.54	74.75	71.99	71.61	71.99	71.71
XGBoost	77.55	78.29	77.55	77.69	74.54	75.95	74.54	74.95	74.07	73.97	74.07	73.99
LightGBM	79.17	79.60	79.17	79.16	77.78	78.63	77.78	78.06	76.62	76.44	76.62	76.48
MLP	71.76	73.70	71.76	72.05	75.92	76.17	75.92	75.95	71.06	72.33	71.06	71.38
DNN	81.94	81.80	81.94	81.80	80.78	82.07	80.78	81.21	82.17	82.84	82.17	82.29
BiLSTM	73.61	74.04	73.61	73.51	73.84	73.20	73.84	73.24	76.62	77.03	76.62	76.48

D. Comparative Analysis

The results show that embedding choice and classifier architecture interact strongly. Longformer embeddings were particularly effective with nonlinear ensemble models such as Random Forest, XGBoost, and LightGBM. BigBird features worked relatively well with linear classifiers and MLP, while BERT was particularly effective for the DNN and BiLSTM.

Overall, deep learning models generally outperformed conventional ML models when using transformer embeddings, although LightGBM remained a competitive and computationally efficient option. The DNN with BERT provided the best embedding-based student model, whereas Longformer fine-tuning produced the highest overall accuracy.

V. DISCUSSION

The experimental findings indicate that contextual transformer representations are useful for distinguishing politically nuanced categories such as Official News, Government Claim, Propaganda, and Opinion. Longformer's local and global attention mechanisms were particularly useful for longer news articles. BERT also produced a strong representation and paired especially well with the DNN.

The relatively lower BigBird result should be interpreted in the context of the experimental configuration. Although BigBird is designed for long sequences using sparse attention, the study limited all models to a 512-token maximum sequence length and used full self-attention for comparability. This reduced the potential computational advantage of its sparse attention design.

The combination of text augmentation, class-balanced weighting, and class-balanced focal loss produced relatively close precision, recall, and F1 values across the evaluated models, indicating that the class imbalance strategy helped avoid excessive majority-class bias.

The study also highlights the practical value of knowledge distillation. Using GPT-4o Mini as a teacher allowed the creation of a labeled dataset with rationales and confidence scores, while the student models offered substantially smaller computational footprints. This is important for applications such as media monitoring, misinformation analysis, and crisis-response systems.

VI. CONCLUSION AND FUTURE WORK

This study developed a teacher–student framework for automated classification of Pakistan–India conflict discourse during April–May 2025. A manually collected dataset of 1,592 articles from 13 Pakistani, Indian, and international mainstream sources was annotated by GPT-4o Mini into four content-framing categories. Llama 3.1 8B Instant was excluded from the final analysis because of inconsistent label generation.

Three transformer representations were evaluated with five ML and three DL student models. Longformer fine-tuning achieved the best overall performance with 82.64% accuracy and 82.75% F1-score. LightGBM was the strongest conventional ML model with 79.17% accuracy, while the BERT-based DNN achieved 82.17% accuracy and 82.29% F1-score, demonstrating a strong lower-cost alternative to full transformer fine-tuning.

Future work should extend the framework to Urdu, Hindi, and Roman Urdu; increase temporal and source diversity; and evaluate robustness under rapidly evolving and adversarial real-world conflict conditions.

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REFERENCES

[1] M. Goossens and F. Mittelbach, "The emperical analysis of gooslings algorithm," in *Proc. 6th Int. Conf. Algorithmic Theory*, pp. 330–331, 2006.

[2] X. Chen, X. Huang, Q. Gao, L. Huang, and G. Liu, "Enhancing text-centric fake news detection via external knowledge distillation from LLMs," *Neural Networks*, vol. 187, p. 107377, 2025.

[3] D. K. Sharma and S. Garg, "IFND: A benchmark dataset for fake news detection," *Complex & Intelligent Systems*, vol. 9, no. 3, pp. 2843–2863, 2023.

[4] L. Hu, Z. Zhao, W. Qi, X. Song, and L. Nie, "Multimodal matching-aware co-attention networks with mutual knowledge distillation for fake news detection," *Information Sciences*, vol. 664, p. 120310, 2024.

[5] M. Kazmi, S. Nauman, S. A. Rauf, S. Ali, A. Daud, and B. Alshemaimri, "Fake news detection on Kashmir issue using machine learning techniques," *International Journal of Advanced Computer Science & Applications*, vol. 16, no. 6, 2025.

[6] B. Liu, A. Wang, and C. Xia, "Interpretable Chinese fake news detection with chain-of-thought and in-context learning," *IEEE Access*, 2025.

[7] S. Prasomphan, "Enhance social network bullying detection using multi-teacher knowledge distillation with XGBoost classifier," *IEEE Access*, 2025.

[8] L. Abualigah, Y. Y. Al-Ajlouni, M. S. Daoud, M. Altalhi, and H. Migdady, "Fake news detection using recurrent neural network based on bidirectional LSTM and GloVe," *Social Network Analysis and Mining*, vol. 14, no. 1, p. 40, 2024.

[9] E. Alsuwat and H. Alsuwat, "An improved multi-modal framework for fake news detection using NLP and Bi-LSTM," *The Journal of Supercomputing*, vol. 81, no. 1, p. 177, 2025.

[10] P. Malhotra and S. K. Malik, "Fake news detection using ensemble techniques," *Multimedia Tools and Applications*, vol. 83, no. 14, pp. 42037–42062, 2024.

[11] M. Mostafa, A. S. Almogren, M. Al-Qurishi, and M. Alrubaian, "Modality deep-learning frameworks for fake news detection on social networks: A systematic literature review," *ACM Computing Surveys*, vol. 57, no. 3, pp. 1–50, 2024.

[12] B. Lakzaei, M. Haghir Chehreghani, and A. Bagheri, "Disinformation detection using graph neural networks: A survey," *Artificial Intelligence Review*, vol. 57, no. 3, p. 52, 2024.

[13] S. Harris, H. J. Hadi, N. Ahmad, and M. A. Alshara, "Multi-domain Urdu fake news detection using pre-trained ensemble model," *Scientific Reports*, vol. 15, no. 1, p. 8705, 2025.

[14] S. Qin and M. Zhang, "Boosting generalization of fine-tuning BERT for fake news detection," *Information Processing & Management*, vol. 61, no. 4, p. 103745, 2024.

[15] B. S. Alqadi, S. A. Alsuhibany, S. N. Yousafzai, S. Alzu'bi, D. M. Alsekait, and D. S. AbdElminaam, "Transfer learning driven fake news detection and classification using large language models," *Scientific Reports*, vol. 15, no. 1, p. 28490, 2025.

[16] T. Hossain, A.-R. Islam, M. H. K. Mehedi, A. A. Rasel, M. Abdullah-AL-Wadud, and J. Uddin, "Correction: Banglanewsclassifier: A machine learning approach for news classification in Bangla newspapers using hybrid stacking classifiers," *PLOS ONE*, vol. 20, no. 9, p. e0332710, 2025.

[17] Y. Gu, Z. Tong, I. Castro, S. Wu, and G. Tyson, "Multi-MLLM knowledge distillation for out-of-context news detection," arXiv preprint arXiv:2505.22517, 2025.

[18] M. Mimura and T. Ishimaru, "Analyzing common lexical features of fake news using multi-head attention weights," *Internet of Things*, vol. 28, p. 101409, 2024.

[19] X. Guo, Q. Liu, Y. Hu, and H. Liu, "MDCNN: Multi-teacher distillation-based CNN for news text classification," *IEEE Access*, 2025.

[20] A. Kishwar and A. Zafar, "Fake news detection on Pakistani news using machine learning and deep learning," *Expert Systems with Applications*, vol. 211, p. 118558, 2023.